Project I: Why use multivariate data analysis?

Last updated on August 18, 2026

Due Dates

1. **November 4**: Project report due (pdf format).

Overview

For this project, you will work independently to develop a brief case study that compares univariate and multivariate methods to answer the question: *Why use multivariate data analysis?* This is a broad question, so you will want to narrow your focus by identifying one aspect of multivariate data analysis to study. For example, some possible questions to target include:

1. How does inference using MANOVA compare with using multiple univariate ANOVAs?
2. How does adding additional response variables affect the performance of different classification or clustering methods?
3. What happens if you have dependent response observations (ex. spatial or temporal dependence)? How does accounting for the correlation between observations affect the results of regression analysis?

The most successful projects will be those that 1) clearly use simulation or real data analysis to illustrate the benefits of multivariate methods compared with standard univariate approaches or 2) identify situations in which multivariate methods may not offer much additional benefit.

Submission Requirements

Report

Develop a 5-7 page PDF report with the following information:

a. **Title, Author, and Date.** The title should be concise, descriptive, and specific.

b. **Introduction.** The introduction briefly describes to the reader 1) the key research question or knowledge gap that is addressed by the study; 2) the data used for this study; 3) the methods used to investigate the research question; and 4) the key findings and why they are important. Three to four paragraphs should be sufficient.
c. **Data.** This section should 1) describe the observational units of the dataset; 2) introduce all relevant variables; 3) describe any relevant details about how the data were generated or collected (and if necessary, transformed); and 4) highlight any critical characteristics of the data, possibly by using visualizations or summary tables.
d. **Methods.** The paper should present the univariate and multivariate methods used in your analysis clearly using mathematical notation and plain English explanations. Every symbol and component of the models should be introduced and described. Your target reader should be a peer in this class. Focus on 1) clearly explaining the key distinctions between the univariate/multivariate methods; 2) explaining how you will compare the methods using your data; and 3) describing what you expect to see in your results (and why).
e. **Results.** The results section should describe the fitted models and the results of analysis. This section should include relevant tables, graphs, and quantitative results, as well as plain English explanations of all of these components, but should not include raw R output. Clearly explain what the results show and interpret them in the context of your research question.
f. **Discussion.** In the final section of the paper, briefly summarize your key findings, discuss potential weaknesses of your study, and identify potential improvements and extensions or future areas of related research.
g. **References.** The text should include in-text citations as well as a reference list for all sources, including any data, software, and relevant literature. You are advised to use BiBTeX for compling your references. All citations should be properly formatted according to an appropriate style guide.
h. **Reproducible workflow.** Create a GitHub repository containing a project folder and appropriate sub-folders. This repository should contain all of your code for your analysis and your final report, **but should not contain any raw or cleaned data from an external source (your collected/simulated data is fine)**. You may use this starter template.

Rubric

See the rubric(s) on Canvas for details, but please note:

1. There should be no R code or raw R output in your submitted PDF report.
2. The paper should be clearly written and appropriate for a target audience of your peers and other undergraduate/graduate students. You should use mathematical symbols and explain them using plain language. You should explain all statistical methods and terms.
3. The project should include a descriptive README file including the author(s) and date of submission and a description of the structure of the project folder. The README

should also include a statement about whether external resources (including LLMs) were used for the project (and if they were, how they were used). The text of the paper and code for the project should be produced entirely by you. The README should also include a statement about where the data is obtained and (if applicable) the license for the dataset. **If either statement is missing, the final grade for this project will be a zero**.

4. All figures, graphs, and tables should be appropriately sized, legible, and free from visual clutter. Clear captions should be provided.